INCH-POUND
A-A-54935
24 December 1993

COMMERCIAL ITEM DESCRIPTION

FREEZER, LABORATORY

(115/230 Volt)

The General Services Administration has authorized the use of this Commercial Item Description.

This Commercial Item Description covers a chest freezer for all types of research and laboratory use.

Salient Characteristics:

The unit shall be a laboratory freezer to be used for all types of research or laboratory use (e.g., viruses, vaccines, pathology specimens, biologicals, blood banking), and wherever any low temperatures are required. The freezer shall have a temperature operating range of plus 10°C to minus 45°C which shall be controlled by a built-in, adjustable, automatic thermostat. A built-in thermometer shall be supplied to indicate the temperature within the storage area.

The refrigeration unit shall consist of a compressor, condenser, drier and evaporator. The refrigerant to effect the cooling may be a halogen compound; however, Class I Ozone Depleting Chemicals (ODC's) are prohibited from being used in this item. The compressor shall be hermetically sealed and shall be rated for 1/2 H.P. The compressor motor shall be permanently lubricated to preclude the necessity for oiling.

AMSC N/A FSC 4110
DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.

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The outside dimensions of the cabinet, including casters, shall be 32-17/32" long by 22-13/32" wide by 43-3/16" high. The dimensions of the storage area shall be 26-1/2" long by 16-1/2" wide by 20" high. The storage area shall have four mounting holes on each end for attaching work shelves. For easy mobility, the cabinet shall be mounted on four 3" lock-type swivel casters with flat plate mountings.

The outer shell of the cabinet shall be fabricated of suitable protected corrosion and rust resistant sheet steel at least 1/32" thick. Commercially accepted practice shall be used for cleaning and priming prior to finishing with paint or lacquer. Finish shall provide a pleasing professional appearance and shall be free of gaps, scratches, chips, flaking and/or other imperfections.

The lining of the storage area shall be fabricated of 1100 aluminum sheet, at least 0.060" thick, and a baked, clear acrylic finish.

The cabinet shall be designed for the top loading of specimens into the storage area. The top of the cabinet shall have a hinged lid. The hinges shall be spring loaded for the self-raising of the lid and for holding the lid open. Three inches of urethane insulation shall surround the storage area on all sides, bottom, and within the top lid. A one piece rubber gasket shall be provided permanently attached to the inner edge of the insulated lid. The lid shall have a latch that can be padlocked for safety and privacy.

Performance. The freezer shall be capable of the following:

a. The freezer shall be soaked-out with cover open and without load, in an ambient temperature between 15°C and 35°C, for a minimum of 8 hours. The freezer shall be operated without load at an ambient temperature between 15°C and 35°C. The freezer shall be in operation for a total of 8 hours in two stages of 4 hours each. One stage shall be run with the thermostat set at minus 40°C. The other stage shall be run with the thermostat set at plus 5°C. The thermometer readings shall be recorded for a minimum of 5 cycles.

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b. The lid shall be closed on a narrow, thin sheet of paper at consecutive points around the position of the gasket. At each position, the sheet shall be pulled out from under the gasket. The proper sealing of the gasket shall be indicated by a slight tug when removing the sheet of paper. If there is no tug, the gasket shall be considered having failed the test.

c. A General Electric Type H-2 Halogen Leak Detector, or equal, shall be used to detect the presence of leaking gas from the refrigeration tubing and joints. The probe of the detector shall be passed over the tubing, joints, gaskets, and welds.

An aluminum storage rack shall be supplied which can be lifted in and out of the storage area. The storage rack shall be perforated to allow air circulation and shall be equipped with handles to facilitate handling. The storage rack shall have two shelves 8-1/4" apart. Each rack shelf shall be 15" by 11-3/4".

A removable access panel shall be provided and located on the cabinet to give accessibility to the electrical and refrigeration components.

The cabinet finish shall be hard, smooth, and highly resistant to corrosion, abrasion, chipping, and cracking.

The laboratory freezer shall be designed for operation on 115/230 volt, 50/60 Hz, AC. A variation of plus or minus 10 percent shall not affect the operation of the freezer. The power consumption of the freezer shall not exceed 750 watts.

A seven foot power cord shall be attached to the cabinet. The cord shall consist of 3-14AWG conductor wire, with one lead grounded. The cord shall be terminated with a lock-in electrical plug to avoid accidental unplugging. A two conductor plug adapter with a grounding pigtail shall be supplied for those installations without grounded receptacles.

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A permanently affixed nameplate shall be attached to the cabinet. The nameplate shall include the following:

1. National Stock Number
2. Type of model number
3. Serial number
4. Name or registered trademark of the manufacturer
5. Federal Supply Code for manufacturer
6. Contract or purchase order number
7. Quarter and year of manufacture
8. Electrical ratings
9. U.S.A.

Marking secured by adhesives shall not be allowed.

The freezer shall conform to UL 471, Standard for Commercial Refrigerators and Freezers.

The contractor shall furnish manuals, handbooks and/or brochures containing complete operation, installation and service/maintenance instructions (including pictures of illustrations, as necessary) with complete schematics and wiring diagrams. All manuals, handbooks, and/or brochures shall be written in the English language and all schematics and wiring diagrams shall use American electrical and electronics symbols. The manual shall include electrical data and connection diagrams for all applicable utilities. The instructions shall also contain a complete list of all replaceable parts showing part number, name and quantity required. All service notes, service memos, etc., pertaining to the equipment and issued subsequent to the printing of the original instructions shall also be included. These instructions shall be, as a minimum, that furnished to service engineers (or servicemen) that normally install and service the equipment for the company or distributor. When the system being procured includes, or will operate ancillary equipment, the service data shall include complete instructions and drawings which show interfacing of all system components.

Two complete sets of the above information shall be packaged with each unit of the required equipment. Where multiple quantities of the same unit are to be delivered to the same ultimate consignee, then only a total of two service manuals need to be supplied to each consignee. However, two operational manuals still need to be supplied with each unit.

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Workmanship. The laboratory freezer shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Each (EA). One laboratory freezer of each voltage, as specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, with an AQL of 1 (percent defective) and an inspection level of II. The unit of product for examination purposes shall be one laboratory freezer of each voltage.

Sampling for tests. Sampling for tests, including dimensional tests, shall be conducted in accordance with MIL-STD-105, with an AQL of 1 (percent defective) and an inspection level of II. The unit of product for test purposes shall be one laboratory freezer of each voltage.

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Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics. Tests shall include, but shall not be limited to, the following:

a. Tests in accordance with UL 471. UL listing mark or a certified test report from an independent laboratory acceptable to the Government shall be accepted as conforming with UL 471.

The freezer shall perform as specified in the Salient Characteristics section, paragraph "a" under Performance.

b. Performance test. The accuracy of the thermostat shall be determined from the recorded thermometer reading. The recorded thermometer readings at equilibrium shall be within plus or minus 5°C of the thermostatic settings.

c. Sealing of lid gasket. The proper sealing of the gasket shall be indicated by a slight tug when removing the sheets of thin paper. If there is no tug, the gasket shall be considered having failed the test.

d. Gas leakage. The unit shall fail if the General Electric Type H-2 Halogen Leak Detector Meter indicates leakage of gas on the most sensitive meter range.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

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Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description, and that the product conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug, and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug, and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

Ordering data. Intermediate/exterior package quantities, labeling, and marking must be specified in the contract and/or order.

NOTE: The following National Stock Number is covered by this document:

4110-01-375-6517

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MILITARY INTERESTS:

Custodians:

Air Force-03
Army-MD
Navy-MS

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

FDA-MPQAS
USPHS
VA-OSS

Project Number: 4110-0481
Location: FREEZER1D-CID

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	A-A-54935	24 December 1993

3. DOCUMENT TITLE: FREEZER, LABORATORY (115/230 Volt)

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
DEFENSE PERSONNEL SUPPORT CENTER ATTN: DPSC-MSE	(1) Commercial AC 215-737-2870	(2) AUTOVON AV 444-2870
c. ADDRESS (Include Zip Code) 2800 SO. 20TH STREET PHILADELPHIA, PA 19145	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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